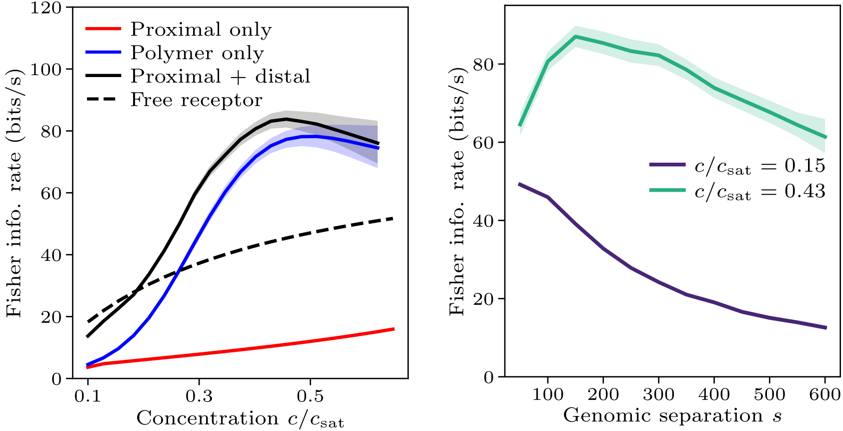
$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t} \left\langle \frac{\partial^2 \mathcal{L}(c)}{(\partial \ln c)^2} \right\rangle_c$$

$$\mathcal{L}(c;t) = \ln P(\,\sqcap\!\sqcap\!\!\!\sqcap\!\sqcap\!\sqcap\!\sqcap\!\sqcap\,|c)$$

$$\dot{\mathcal{F}}(c) \approx \dot{\mathcal{F}}_{local}(c) + \dot{\mathcal{F}}_{\mathrm{D}NA}(c)$$


Fisher info. rate (bits/s)
120
100
80
60
40
20
0
0.1
0.3
0.5
Concentration c/c_{sat}
Proximal only
Polymer only
Proximal + distal
Free receptor
Fisher info. rate (bits/s)
80
60
40
20
0
100
200
300
400
500
600
Genomic separation s
$c/c_{\mathrm{sat}} = 0.15$
$c/c_{\mathrm{sat}} = 0.43$